## Extended temporomandibular joint prostheses: a retrospective analysis of feasibility, outcomes, and complications.

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Prostheses for extended total temporomandibular joint replacement (eTJR) include modifications to the traditional alloplastic fossa-condyle joint that extend to adjacent bone defects. The aim of this retrospective study was to assess the feasibility, postoperative complications, and functional and aesthetic outcomes after eTJR. Patients aged ≥18 years undergoing eTJR between 2013 and 2022 were included. Data recorded were age, sex, comorbidities, indication for eTJR, prosthesis brand, classification, concomitant surgical procedures, postoperative complications, maximum inter-incisal opening (MIO), pain, quality of life (QoL), and aesthetic outcome. Twenty-five patients (mean age 40 years), with a total of 30 joint prostheses, were included. Over a median follow-up of 42 months, there was a significant improvement in MIO in patients with reduced mouth opening at baseline ($P = 0.003$), as well as in pain ($P = 0.007$) and QoL ($P = 0.004$). Both patients and surgeons judged facial appearance as improved or unchanged in 88% of cases. Postoperative complications included permanent trigeminal nerve hypoesthesia (44%), permanent facial nerve dysfunction (35%), infection (8%), salivary leak (4%), and lingual nerve impairment (4%). The findings suggest that eTJR is a safe and effective treatment for temporomandibular joint deficits extending to adjacent structures, yielding satisfactory functional and aesthetic outcomes.

## Can growing patients with end-stage TMJ pathology be successfully treated with alloplastic temporomandibular joint reconstruction? - A systematic review.

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The use of alloplastic total temporomandibular joint reconstruction (TMJR) in growing patients is controversial, mainly due to immature elements of the craniomaxillofacial skeleton. The aim of this systematic review was to evaluate the use of alloplastic TMJR in growing patients, focusing on the patient's clinical presentation, surgical and medical history and efficacy of alloplastic TMJR implantation. The literature search strategy was based on the Population, Intervention, Comparator, Outcomes and Study type (PICOS) framework. We searched Pubmed, Google Scholar, Dimension, Web of Science, X-mol, Semantic Scholar and Embase to January 2023, without any restriction on the type of publication reporting alloplastic TMJR in growing patients (age ≤ 18 years for boys and age ≤ 15 years for girls). A total of 15 studies (case reports: 09, case series: 02, cohort studies: 04) met the inclusion criteria, documenting 73 patients of growing age from 07 countries. Thirty-eight (~ 52%) cases were female. The mean ± SD (range) age and follow-up of patients in all studies was 13.1 ± 3.2 (0-17) years and 34.3 ± 21.5 (7-96) months, respectively. A total of 22 (30%) patients were implanted with bilateral alloplastic TMJR. Over half of the studies (n = 10) were published in the last 3 years. All patients underwent multiple surgeries prior to implantation of alloplastic TMJR. In extreme cases, patients underwent a total of 17 surgeries. Different types of studies reporting inconsistent variables restricted our ability to perform quality assessment measures for evidence building. Clinical experience with alloplastic TMJR in growing patients is limited to cases showing poor prognosis with other types of reconstruction. Nevertheless, studies show promising results for the use of alloplastic TMJR in growing patients, highlighting the need for well-controlled prospective studies with long-term follow-up.

## Temporomandibular joint disorders: artificial joint replacements and future research needs.

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Temporomandibular disorders (TMD) affect a significant section of the American population. According to the National Institutes of Health (NIH), an estimated 3% to 5% of Americans suffer from temporomandibular disorders. Majority of TMDs can be treated by conservative methods, albeit surgical interventions are indicated for some pathological/clinical conditions (according to the American Association of Temporomandibular Joint Surgeons). A distinction can be made between the TMD and the TMJ disease/dysfunction. In the case of TMD, it most correctly relates to a neuromuscular type of problem in the general area of the Temporomandibular Joint (TMJ), but may not intrinsically be related to the TM joint itself. However, TMJ disease/dysfunction relates to a true joint type of pathology, which ultimately may lead to the use of either a partial or a total joint reconstruction. Degeneration of the meniscus and the actual bony portions of the joint is often seen in cases of true TMJ pathology. The use of custom made or predesigned partial and/or total artificial TMJ replacement remains one of the surgical alternatives for treating various TMJ diseases when other conservative treatments fail. The purpose of this article is to provide an overview of currently existing TMJ implants and to discuss relevant issues, including some of the challenges and current needs facing TMJ research. Additionally, this article describes some of the currently available TMJ surgical procedures, the use of autografts and alloplastic materials (with primary emphasis on TMJ implants) for TMJ reconstruction, and some suggestions for future research. Past incidents of clinical failures of TMJ implants have been attributed to several reasons. The important factors that are responsible for the past failures include lack of a sound scientific approach and inadequate basic research to study the underlying causes for pathologies and their remedies of the TMJ. All these previously mentioned concerns have prompted current biomedical researchers to address many overlooked issues in relation to studying temporomandibular disorders and TMJ implants. Future research should be aimed toward addressing TMJ problems with a multidisciplinary approach, emphasizing simultaneous involvement from clinical (i.e., dentists and physicians) and nonclinical personnel (i.e., oral biologists, bioengineers, biostatisticians, physiotherapists, and pain management specialists).

## Advancements in temporomandibular joint total joint replacements (TMJR).

*PMID: 31168422 | Indexed in PubMed*

The goal of this paper is to review the advantages and disadvantages of the various treatment options of temporomandibular joint (TMJ) total joint replacement (TJR). TMJ articles published within the last 20 years were reviewed to collect the information on non-invasive and invasive TMD treatment methods. Recent technological advancements helped the evolution of treatment methods and offered significant value to TMD patients and surgeons. Considering the TMD levels, the therapeutic procedures can involve general health examiniations, physical therapy, medication, oral rehabilation or as an end stage clinical invention, temporomandibular joint replacement. In fact when intra-articular TMD is present, the effective treatment method appears to be TJR. However, concern for infection, material hypersensitivity, device longevity and screws loosening issues still exists. Further combined research utilizing the knowledge and expertise of, surgeons, material scientists, and bioengineers is needed for the development of improved TMD therapeutic treatment.

## Application of the rapid prototyping technique to design a customized temporomandibular joint used to treat temporomandibular ankylosis.

*PMID: 19881026 | Indexed in PubMed*

Anthropometric variations in humans make it difficult to replace a temporomandibular joint (TMJ), successfully using a standard "one-size-fits-all" prosthesis. The case report presents a unique concept of total TMJ replacement with customized and modified TMJ prosthesis, which is cost-effective and provides the best fit for the patient. The process involved in designing and modifications over the existing prosthesis are also described. A 12-year- old female who presented for treatment of left unilateral TMJ ankylosis underwent the surgery for total TMJ replacement. A three-dimensional computed tomography (CT) scan suggested features of bony ankylosis of left TMJ. CT images were converted to a sterolithographic model using CAD software and a rapid prototyping machine. A process of rapid manufacturing was then used to manufacture the customized prosthesis. Postoperative recovery was uneventful, with an improvement in mouth opening of 3.5 cm and painless jaw movements. Three years postsurgery, the patient is pain-free, has a mouth opening of about 4.0 cm and enjoys a normal diet. The postoperative radiographs concur with the excellent clinical results. The use of CAD/CAM technique to design the custom-made prosthesis, using orthopaedically proven structural materials, significantly improves the predictability and success rates of TMJ replacement surgery.

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## Temporomandibular joint replacement.

*PMID: 7613719 | Indexed in PubMed*

Temporomandibular joint replacement is a small area in maxillofacial surgery but of great importance. The treatment of severe temporomandibular disease with prosthetic joints can produce enormous benefit to patients and the current status of this technique is discussed here.

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## A Minimally Invasive Approach for Temporomandibular Joint Replacement: A Pilot Study.

*PMID: 34453907 | Indexed in PubMed*

Temporomandibular joint replacement (TJR) with an alloplastic (metal/ultra-high-molecular-weight polyethylene) device has proven to be a successful and predictable procedure. This paper describes a novel technique for performing TJR with an endaural incision alone. The technique we are describing uses only an endaural incision with supplemental trocar incision(s), to perform a TJR. There were 4 patients for a total of 8 temporomandibular joints that were selected. All 4 patients were assessed immediately following surgery, on postoperative days 1 and 7 and at 6 months following surgery. Maximal interincisal opening and subjective variables were assessed at each of the time points. Additionally, the total operative time was measured and compared to a previous age and diagnosis matched control group using the traditional 2 incisions TJR.There were 3 females and 1 male (ages 19-67) who underwent TJR with an endaural incision alone. There were 4 females (ages 19-68) who underwent traditional TJR surgery. None of the patients in either group had major complications and all patients were discharged on postoperative day 1. All patients in the endaural incision alone group had increased maximal interincisal opening and reported a quicker subjective decrease in pain and disability following surgery with less average time in the operating room. However, all patients in the endaural incision alone group had CN VII weakness that lasted longer than those in the traditional TJR group.The minimally invasive approach for TJR was successful in the present pilot study and could be used in specific situations to decrease the morbidity associated with additional incisions for this procedure. Ultimately, the endaural only incision approach offers promising outcomes for future patients undergoing temporomandibular joints TJR in the right patient population.

## Extended Total Temporomandibular Joint Replacement - A Feasible Option for Functional and Aesthetic Reconstruction of Mandibular Defects Involving the Temporomandibular Joint.

*PMID: 37975029 | Indexed in PubMed*

Study design: Retrospective case series. Objective: Alloplastic temporomandibular joint replacement has been established as a standard technique for end- stage temporomandibular (TMJ) pathologies. Joint replacement when there are extensive mandibular defects remains a challenging clinical problem. Custom-made extended temporomandibular joint replacement is a feasible option but there is limited information about this emerging technique. Methods: Included were all patients undergoing extended TMJ-replacements (TMJe), all operatrions were carried out by the senior author. Surgical technique was either single stage or two stage protocol. Surgical details and pitfalls and outcome of more than 2 years follow-up with reference to thirteen including twelve patients were recorded. Results: The most common diagnosis was ameloblastoma of the mandibular ramus. Single stage or two stagge regime were carried out depending on resection requirements and involvement of teeth. Improved mouth opening of more than 30mm was achieved in 10 of 12 patients. One patient with previous TMJ replacement reported temporary weakness of the facial nerve, which resolved after 10 months. Conclusions: The authors suggest a simplified anatomically based single-stage or two-stage regime, with both regimes achieved excellent anatomic reconstruction, facial appearance and function with low surgical morbidity. Custom-made extended temporomandibular joint protheses appear an advanced and reliable solution for reconstruction of combined complex mandibular defects including the temporomandibular joint. If surgical clearance of the pathology can be achieved, a single-stage regime is favoured.

## [Total temporomandibular joint prostheses].

*PMID: 27554487 | Indexed in PubMed*

The temporomandibular joint (TMJ) is probably the most complex human joint. As in all joints, its prosthetic replacement may be indicated in selected cases. Significant advances have been made in the design of TMJ prostheses during the last three decades and the indications have been clarified. The aim of our work was to make an update on the current total TMJ total joint replacement. Indications, contraindications, prosthetic components, advantages, disadvantages, reasons for failure or reoperation, virtual planning and surgical protocol have been exposed.

## Outcomes of open temporomandibular joint surgery following failure to improve after arthroscopy: is there an algorithm for success?

*PMID: 23701829 | Indexed in PubMed*

We reviewed the results of one surgeon's experience of open surgical management of the temporomandibular joint (TMJ) in patients who fail to respond to arthroscopy and aimed to identify groups of patients that may or may not benefit from the intervention. Over a 7-year period (2005-2012) we retrospectively collected data from the medical notes of patients who underwent discectomy, disc plication, eminectomy, eminoplasty, and adhesiolysis, according to the clinical findings for joint pain, restriction, and locking. A total of 22 patients (71%) reported improvement in pain score and 19 (61%) reported an improvement in mouth opening 12 months postoperatively. Overall, 12 patients (39%) ultimately needed TMJ replacement. This group included 5/6 patients in Wilkes' stage IV and 6/15 in stage V, 5/7 patients with a preoperative pain score of 90-100, and half of those with preoperative mouth opening of 20-29 mm (7/14). Open surgical management of the TMJ can benefit patients despite the previous failure of arthroscopy to manage pain, restriction, and locking. Arthroscopy seems to reduce the percentage of patients that need open TMJ surgery, but also the success of subsequent operations compared with previous studies. TMJ replacement is increasingly being done successfully to treat end-stage disease. These results may be used when obtaining a patient's consent for open TMJ surgery, particularly if they are in the groups considered to have a high risk of subsequently requiring a replacement joint.